Vision

Monte Carlo Simulation & Synthetic Data for Autonomous Systems

Powered by UNSEEN ENGINE

The Hidden Danger: "Black Swan" Events

99.9% of miles driven are easy. The 0.1% of unexpected, rare events are what cause accidents and prevent full autonomy.

Data Can't Be Collected

It is physically impossible, dangerous, and economically unviable to collect real-world data on catastrophic edge cases. You cannot film a thousand fatal crashes.

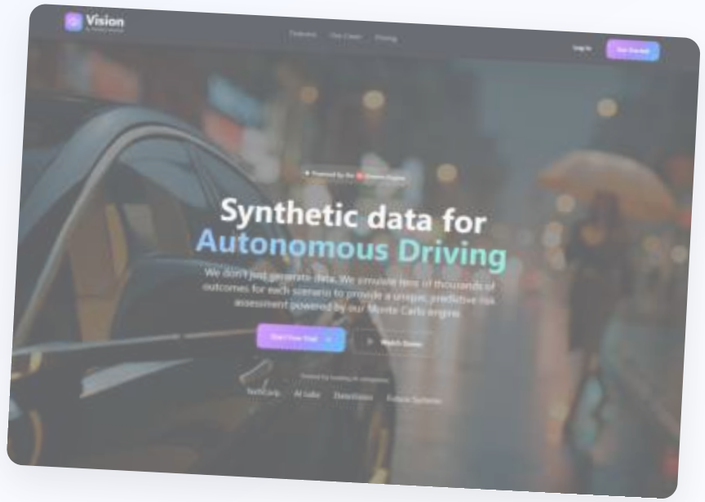
Simulators Can't Imagine

Traditional simulators are limited by human imagination. They can't generate the truly "unseen" scenarios that cause systems to fail in the real world.

AI Lacks Foresight

A standard perception system can say "this is a car," but it cannot quantify the risk or uncertainty in a complex scene. It operates on determinism, not probability.

The Core Problem: The entire industry is bottlenecked by the inability to train and test on the most critical, life-or-death scenarios: the "Black Swans."



Our Answer: Data That Sees the Unseen

We generate high-fidelity synthetic data and enhance it with a powerful, proprietary layer of probabilistic risk analysis.

Core Product: Datasets for Low-Light Conditions

You receive comprehensive, structured datasets focused on the most challenging scenarios: night, rain, fog, and other low-visibility conditions. This is the essential fuel for training robust perception systems.

Unique Feature: The UNSEEN ENGINE Analysis

As a powerful addition, every dataset comes with our proprietary Monte Carlo analysis. We provide a quantitative, probabilistic assessment of risk for each scene, turning simple data into actionable safety insights.

The Two Questions That Matter

We provide quantitative answers to the most critical safety questions.

1. What is the probability of avoiding a collision?

Even if the AI controller performs perfectly, what is the chance that the laws of physics make a collision unavoidable? We calculate this probability through thousands of simulations.

2. If a collision occurs, what is the probability it won't be fatal?

We analyze the "heavy tails" of the outcome distribution. This allows us to quantify the likelihood of severe outcomes vs. minor incidents, providing a true measure of risk, not just a pass/fail.

Answering the Critical Questions: Our Process

To answer these critical safety questions, we've built a powerful, three-step process.

1

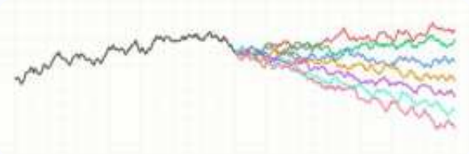
Define Scenario

Specify parameters for the data you need via our web interface or API (e.g., "night, heavy rain, pedestrian appears from behind a bus").

2

Generate & Analyze

Our platform generates a rich, structured dataset with images, depth maps, and segmentation masks, then runs thousands of Monte Carlo simulations.



3

Get Data Package

Download the complete, ready-to-use data package with all assets and the crucial risk metadata for each scene, ready for your pipelines.

One Core Engine, Many Applications

Our powerful simulation engine is a reusable platform, delivering value across different industries.

Unseen Vision
(Automotive)

Road scenarios, sensor data, and vehicle dynamics.

[Future Product]
(e.g., Finance, Robotics)

Domain-specific parameters like portfolios or kinematics.

→

UNSEEN ENGINE

API (FastAPI)
↓
Task Queue (Celery)
↓
Broker (Redis)
↓
Monte Carlo Workers
↓
Database (Postgres)

→

Analyzed Data Package
(The "Black Swan" Output)

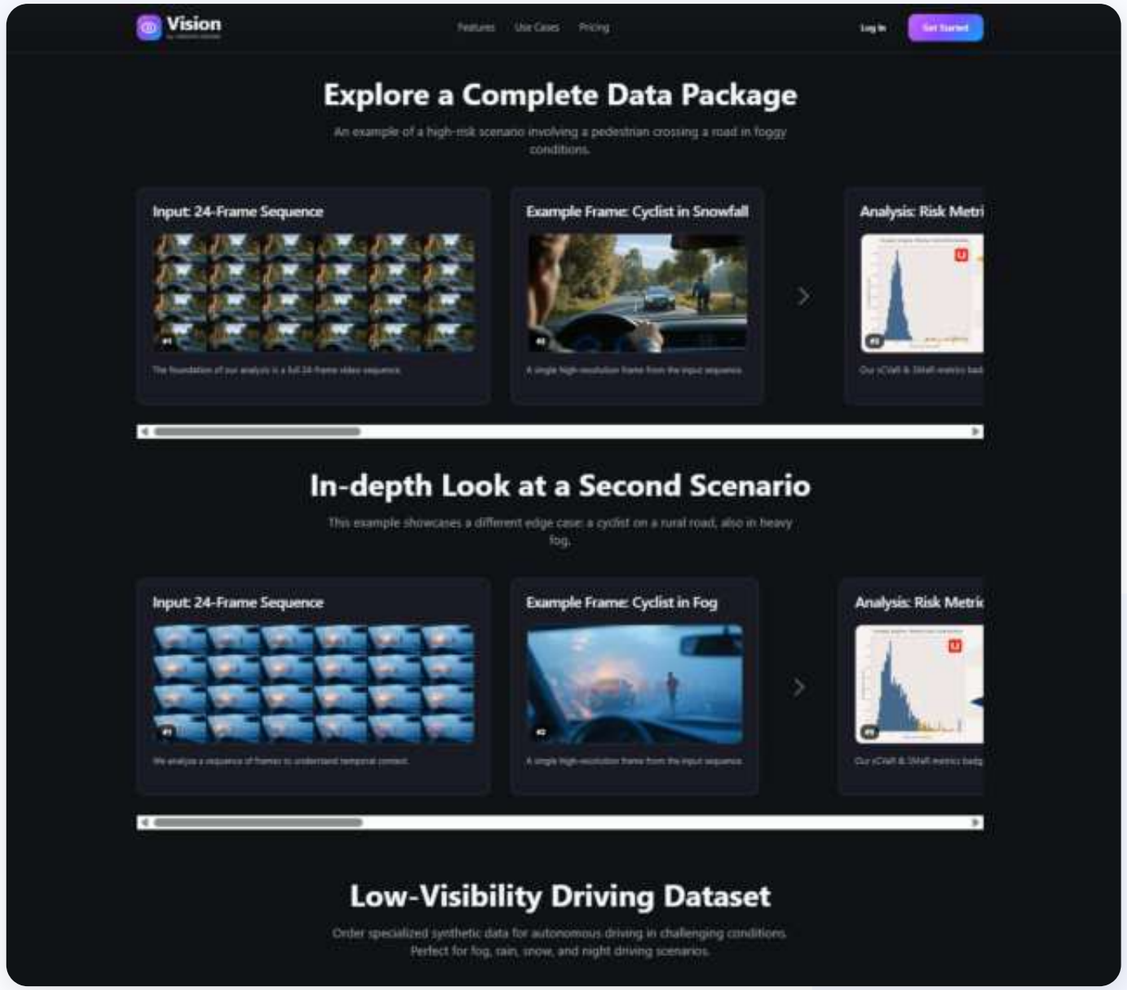
- Vision Data Packages
- sCVaR / SMaR Metrics
- Actionable Safety Insights

The Vision Data Package

You get more than just images. You get a complete, structured, ready-to-use asset.

Standard Package Contents:

- `frames/` Primary RGB driver-view frames
- `depth/` Monocular depth maps (.npy)
- `seg_masks/` Semantic & instance segmentation masks
- `bboxes/` 2D/3D bounding boxes in COCO format
- `tracks.json` Per-instance tracks & 3D positions
- `camera_intrinsics.json` Camera calibration data
- `ego_poses.json` Ego-vehicle poses
- `scene_meta.json` Scene metadata (weather, visibility, etc.)
- `mc/` **The Core Value:** Monte Carlo analysis
  - `mc_config.json` Monte Carlo configuration
  - `mc_results.json` Aggregated Monte Carlo results
- `badge.png / badge.svg` Visual summary badge
- `manifest.json` Top-level manifest
- `provenance/` Data generation provenance



Key Advantages

What makes Unseen Vision different.

Focus on "Black Swans"

We systematically generate rare, high-impact scenarios ("Black Swans") like pedestrians appearing in fog, which are impossible or unethical to collect in the real world.

Quantitative Risk Assessment

We provide proprietary, patent-pending metrics: **sCVaR_α** (severity analysis) and **SMaR_α** (safety margin) to quantify *how likely* and *how severe* a potential collision could be.

Physics-Based Validation

Every scenario undergoes thousands of Monte Carlo simulations to verify physical consistency and answer critical questions about collision probability and outcome severity.

Full Sensor Emulation

Our datasets provide perfect ground truth not just for RGB cameras, but also for depth, segmentation, and emulated LiDAR and RADAR, ensuring comprehensive sensor fusion testing.

Turnkey Data Packages

Receive structured, ready-to-use packages in standard formats (COCO, NPY) that plug directly into your existing MLOps pipelines, saving significant data engineering effort.

On-Demand Generation

Our platform allows flexible configuration of scenario parameters (weather, time of day, event type) via API or web interface, delivering tailored data on demand.

Target Audience & Applications

Who benefits from Vision's powerful data generation.

Automotive OEMs

Validate and certify your AV stack with statistically significant safety data.

Tier 1 & 2 Suppliers

Test and benchmark your perception sensors and control software in challenging scenarios.

AV Startups

Accelerate your development with high-quality, affordable training and validation data on demand.

Insurance & Regulation

Develop new risk models and establish transparent, data-driven safety standards.

Academia & Research

Access vast, affordable datasets for computer vision research and training future engineers.

Robotics & Drones

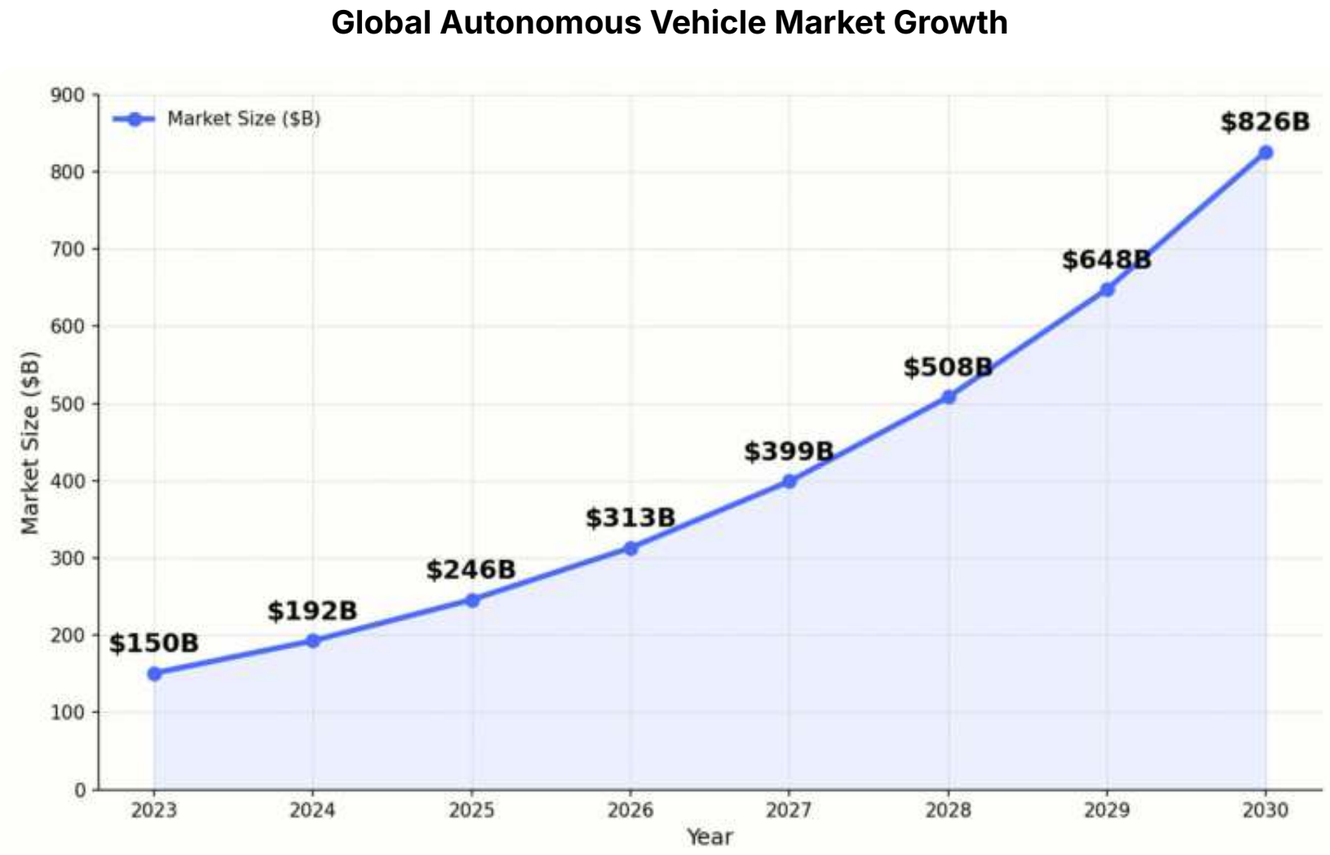
Test navigation & object avoidance models for complex, unstructured environments.

Defense & Aerospace

Create high-fidelity simulations for mission-critical testing and validation of autonomous systems.

Market Opportunity

The Autonomous Vehicle market is massive, but gated by safety validation.



30%+

Market Growth (CAGR)

The AV market is expanding rapidly year-over-year.

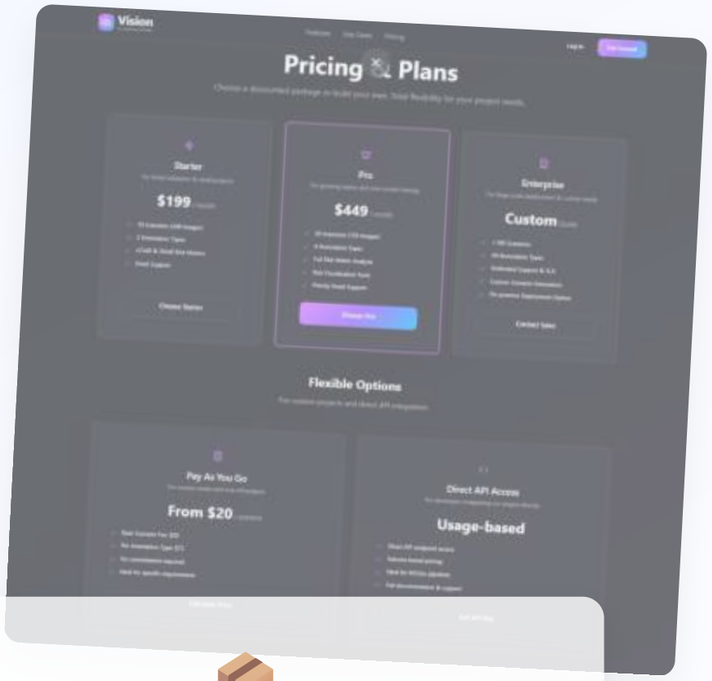
90%

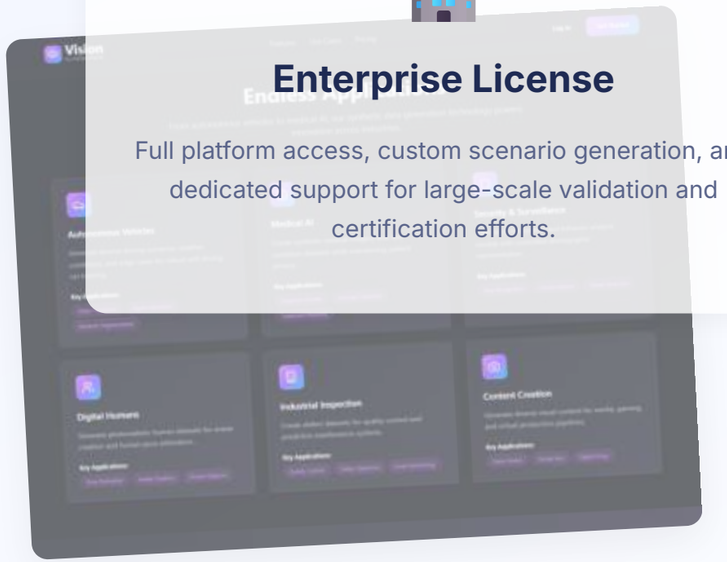
Industry Bottleneck

Verifiable safety and testing is the #1 challenge holding back mass adoption.

Business Model

Flexible and scalable access to our data generation platform.





Enterprise License

Full platform access, custom scenario generation, and dedicated support for large-scale validation and certification efforts.

API & Subscription

For high-volume users who need continuous integration into their MLOps pipelines. Get programmatic access to data generation.

Pay-as-you-Go

Purchase individual, pre-generated or custom-built data packages on demand. Perfect for initial research and smaller projects.

The Ask

Investment Opportunity & Team Expansion

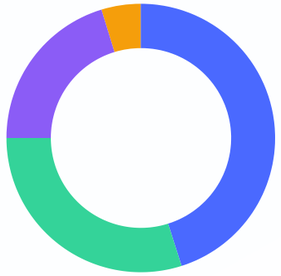
Funding Target

$2.5M

Series A round to accelerate growth and market penetration

Est. Valuation	**$10M**
Target Return	**4.5x ROI**
Break-Even	**18 months**
Equity Offered	**20%**

Use of Funds



- Product Development (45%)
- Team Expansion (30%)
- Marketing & Sales (20%)
- Operations (5%)

Expected Outcomes

Market Expansion
Penetration into 3 new markets with 5x user growth

Platform Enhancement
Complete API ecosystem & B2B white-label solution

Core Team & Key Hires

CEO & Founder A. Turtsevych	**Executive Director** *Open Role*	**BizDev Lead** *Open Role*
Lead ML Engineer *Open Role*	**Data Engineer** *Open Role*	**DevOps/MLOps** *Open Role*

Get in Touch

Let’s build the future of safe autonomy together.

Andrew Turtsevych

CEO & Founder

10+ years in IT with proven leadership, Engineering & Theoretical Physics background.

UNSEEN ENGINE

Trusted by innovators. Ready for the world.

VISION
by UNSEEN ENGINE

Live Prototype
unseenengine.tech

Founder Contact
andrew@unseenengine.tech

Company Profile
linkedin.com/company/unseen-engine

Message Us on LinkedIn

Let's build resilient autonomy — together.

We design datasets to stress-test safety systems.

We quantify rare-event risk so you can prioritize fixes.